# RedstoneLab

Redstone meters and piston previews, exact to the tick

**Minecraft 1.21.11** · Fabric · 0.3.1

by **kyoubitt**



# 1.1 Install

- **Fabric Loader** + **Fabric API** for 1.21.11.
- `redstonelab-0.3.1.jar` (not `-sources`) into `mods`.
- Nothing else needed. Mod Menu optional.

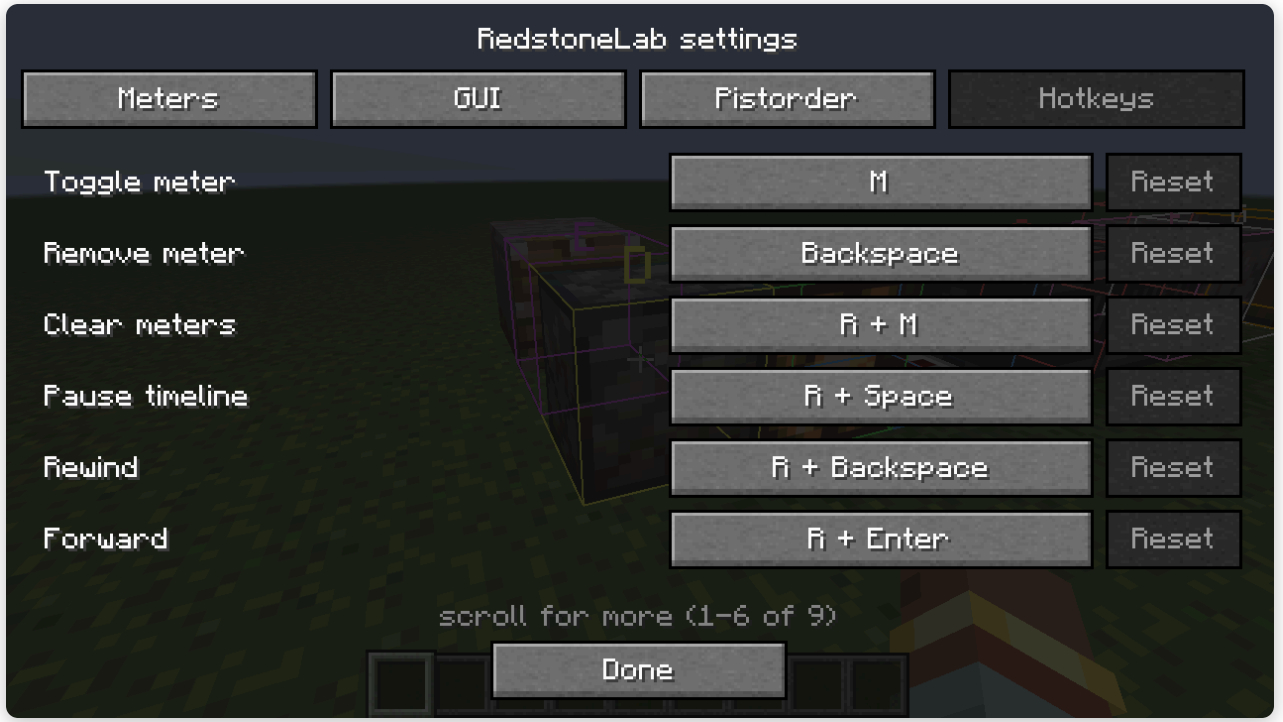
> `github.com/kyoubitt/RedstoneLab/releases`

| On | You get |
|---|---|
| Client only | Piston previews, any server |
| Client + server | Everything, exact ticks |
| Server only | Meters as boss bars / chat, piston previews |

# 1.2 Keys

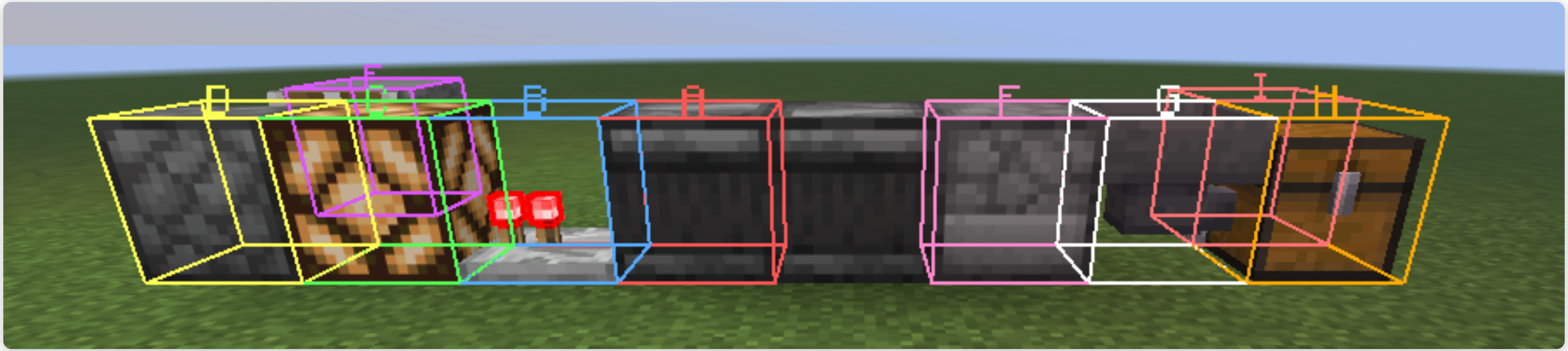
| | |
|---|---|
| `M` | Meter on / off |
| `Backspace` | Remove a meter |
| `R` + `M` | Remove all |
| `Up` | Freeze game / step 1t |
| `Down` | Unfreeze game |
| `R` + `Space` | Freeze timeline |
| `R` + `Backspace` | 1 tick back |
| `R` + `Enter` | 1 tick forward |
| `P` | Hide piston previews |
| `I` | Item labels |
| `R` + `G` | Meter groups |
| `R` + `C` | Settings |

> Change: Settings → Hotkeys, click one, hold the keys, let go. `Esc` unbinds.



# 2.1 Meters

- `M` on a block = meter. `M` again removes it.
- `Backspace`: remove one. `R` + `M`: remove all.
- Any block: dust, repeaters, pistons, observers, droppers, hoppers, chests…
- Letter + colour per meter. Follows blocks moved by pistons.



# 2.2 Timeline




1. **Order**, from A: `A → 2t → B D F G` = B, D, F, G 2t after A
2. Meter letter
3. **Bar** = on, number = ticks
4. Block + on / off lengths

**`6t`** exact **`3t/5t`** alternates **`2t/3t/3t`** pattern **`2t or 3t`** no pattern **`thin line`** powered, not on **`white line`** 0-tick

# 2.3 Per block

| Block | Bar | Text |
|---|---|---|
| Dust, lamp, observer | on | `on 2t · off 4t` |
| Repeater | on | `delay 4t · locked` |
| Torch | lit | `burnt out` (8 flips in 60t) |
| Plain block | powered | `strong 15` lights dust · `weak 13` doesn't |
| Piston | out | `moves every 6t · out 2t` |
| Dropper, dispenser, crafter | fires | `27 Stone · fires every 4t` |
| Hopper | locked | `out every 8t` |
| Chest, barrel… | signal > 0 | `signal 1 (2 at 124)` = 2 at 124 items |
| Comparator | on | `subtract · back 15 · side 14` |
| Target, daylight, sculk, weighted plate | on | `power 7` |

# 2.4 Sub-tick order




1. `R` + `Backspace` / `Enter`: 1 tick back / forward. Back at now it runs again.
2. What happened that tick, **in game order**, by part of the tick.

> `Up` freezes the game (op), again = step 1 tick. `Down` runs it. Block / shape updates: Settings → **Show event marks**.

# 2.5 Commands

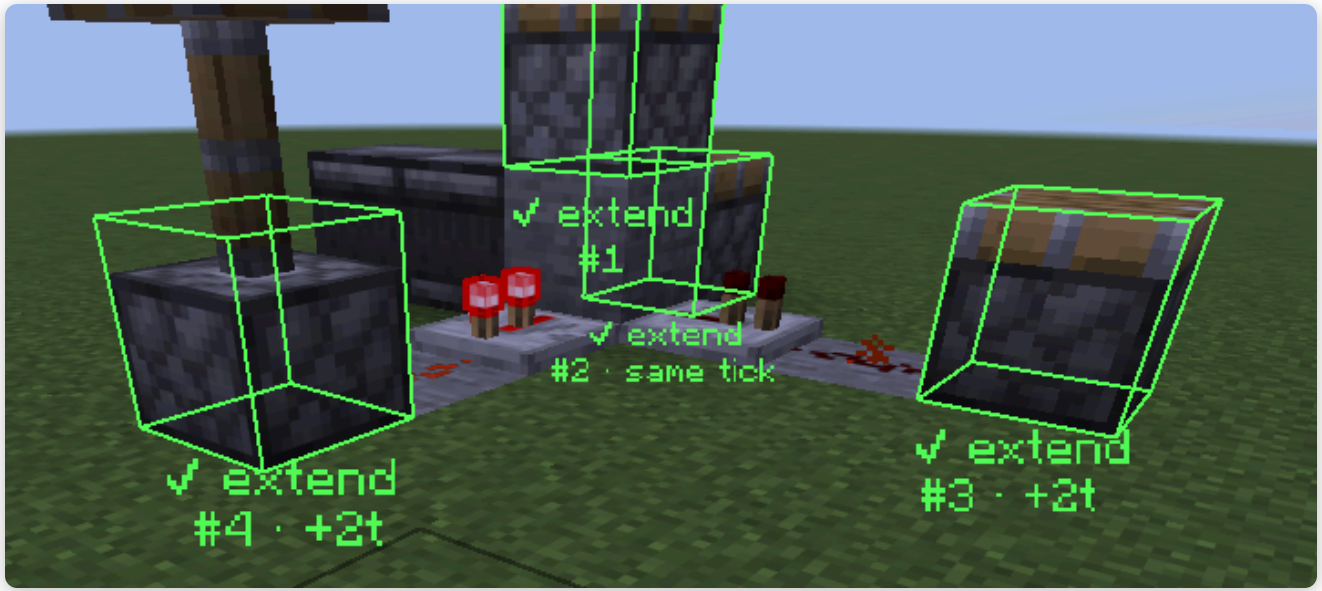
| | |
|---|---|
| `/rlab meter expect A 2 4` | On 2t, off 4t. Wrong = red. |
| `/rlab meter name A clock` | Name for A. |
| `/rlab group <name>` | Saved meter sets. `R` + `G` lists them. |
| `/rlab meter` | Meters in chat (no client mod needed). |
| `/rlab tool` | Right-click tool, no keys needed. |

# 3.1 Piston previews




- **Empty hand, right-click a piston.** Again hides it. `P` hides all.
- ✔ **push 3/12**: numbers = move order. `breaks` = destroyed.
- ✖ **blocked**: `stuck` = the block in the way.
- ✖ **too many**: `13th` = the extra block.
- ⚡ **QC** powered from above.
- ⚡ **BUD** moves on next update.
- Any server: only your client needs it.

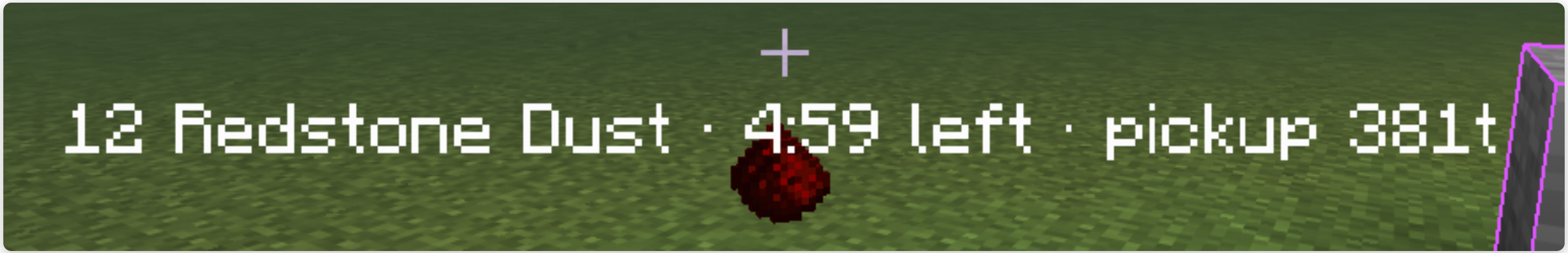
# 3.2 Firing order




- Several shown: order they extend, from the first one shown.
- `+2t` = 2t after the one before.
- `same tick` = same tick, moved after it.
- `own timing` = different speed.

> Exact with the mod on the server. Otherwise lag can shift ticks by 1.

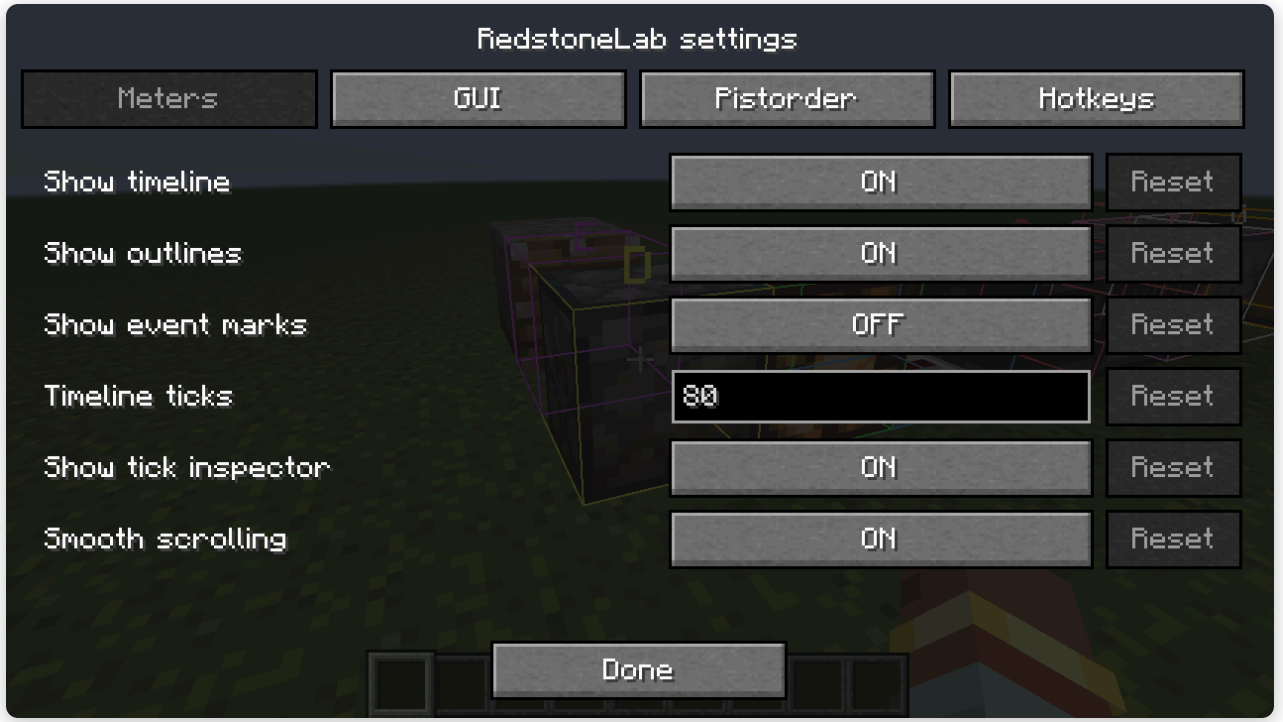
# 3.3 Item labels




- `I`: labels over dropped items, updated every tick.
- `4:59 left` = until it despawns (5 min). Red under 30 s, ticks under 10 s.
- `pickup 381t` = until players / mobs can take it. Hoppers ignore it.
- Nearest 64 within 16 blocks. Needs the mod on the server.

# 4 Settings

- `R` + `C` or Mod Menu.
- **Meters**: what shows, ticks, event marks.
- **GUI**: position, size, background.
- **Pistorder**: landing boxes.
- **Hotkeys**: any key or combo.
- Saves instantly. Hover for help.



# Links

| | |
|---|---|
| Download & source | `github.com/kyoubitt/RedstoneLab` |
| Help in game | `/rlab help` |
| License | LGPL-3.0 |